

Second Terminal Examination - 2082

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET - A

Pass Marks: 30

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Attempt all the questions.

Group 'A'

[11 × 1 = 11]

Rewrite the best alternative of each question in your answer sheet.

- The sum of oxidation numbers of all atoms in $K_4[Fe(CN)_6]$ is
a. +2 b. -1 c. 0 d. +6
- The ability of a given substance to assume two or more crystalline structures is called
a. Polymorphism b. Allotropy c. Isomorphism d. Isomerism
- What will happen to the volume of a bubble of air underwater in a lake, where the temperature is 15°C and the pressure is 1.5 atm, if the bubble then rises to the surface, where the temperature is 25°C and the pressure is 1 atm?
a. Its volume will become greater by a factor of 2.5
b. Its volume will become greater by a factor of 1.6
c. Its volume will become greater by a factor of 1.1
d. Its volume will become smaller by a factor of 0.70
- Since the atomic weights of C, N and O are 12, 14 and 16, respectively, among the following pairs of gases, the pair that will diffuse at the same rate is
a. CO_2 and N_2O b. CO_2 and N_2O_2 c. CO_2 and CO d. CO and NO
- A triple bond consists of:
a. One sigma and one pi bond b. Three sigma bonds
c. One sigma and two pi bonds d. Two sigma and one pi bond.
- Which of the following has both covalent and ionic bonds?
a. H_2O b. NaCl c. KCN d. O_2
- Formula of functional group for aldehyde is
a. $-\text{OH}$ b. $-\text{CHO}$ c. $-\text{COOH}$ d. $-\text{CO}-$
- A fuel having octane number 60 produces the same extent of knocking as produced by fuel mixture containing
a. 60% iso-octane and 40% α -methyl-naphthalene
b. 60% n-heptane and 40% n-hexadecane
c. 40% α -methyl-naphthalene and 60% n-heptane
d. 60% iso-octane and 40% n-heptane
- Tailing of mercury is due to the formation of
a. Mercuric oxide b. Mercurous oxide
c. Mercuric chloride d. Mercurous chloride

10. What happens when sulphur dioxide gas is passed through hydrogen sulphide gas?
 a. SO_2 is oxidized to sulphur b. H_2S is reduced to sulphur
 c. SO_2 is oxidized to H_2SO_4 d. SO_2 is reduced to sulphur
11. The aqueous solution of SO_2 is also known as
 a. Thiosulphurous acid b. Thiosulphuric acid
 c. Hydrosulphuric acid d. Sulphurous acid

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. The electronic concept of redox reaction is based on electron transfer processes. [1+1+3]
 a. Define oxidation and reduction according to the electronic concept.
 b. Define redox reaction.
 c. Balance the following reaction using the oxidation number or ion-electron method.
 $\text{KMnO}_4 + \text{HCl} \longrightarrow \text{KCl} + \text{MnCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$

OR

Michael Faraday investigated the quantitative aspect of electrolysis by measuring the amount of chemical change by a known quantity of electricity. [1+1+3]

- a. State Faraday's first law of electrolysis.
 b. Define ECE.
 c. How long a current of 3 amperes has to be passed through a solution of AgNO_3 to coat a metal surface of 80 cm^2 with a 0.005 cm thick layer? (density of $\text{Ag} = 10.5 \text{ g/cc}$ and atomic weight of $\text{Ag} = 108$)
13. The ideal gas equation helps convert the moles of gas obtained from electrolysis into a measurable volume under specified conditions of temperature and pressure. [3+2]
 a. Derive the ideal gas equation.
 b. Write the differences between real gas and ideal gas.
14. Boyle's law can be verified by plotting different values of volumes of a certain mass of gas under different pressures. [1+1+1+2]
 a. State Boyle's law.
 b. Draw a graph to illustrate Boyle's law, showing the relationship between volume (V) and pressure (P).
 c. Write the applications of Boyle's law.
 d. At what temperature would the volume of a gas be tripled, if the pressure at the same time increases from 650 mm to 750 mm pressure, the gas being at 15°C initially?
15. Molecular formulae of some compounds are given below:
 Boron trifluoride BF_3
 Methyl alcohol CH_3OH
 Sulphuric acid H_2SO_4
 Sodium carbonate Na_2CO_3

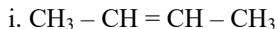
- a. Predict the mode of hybridization of B in BF_3 and also predict the geometry of BF_3 on the basis of hybridization. [2]
- b. Why is methyl alcohol soluble in water? [1]
- c. Draw the Lewis structure of i) Sulphuric acid ii) Sodium carbonate. [2]

16. Homologous series is the series of organic compounds which can be represented by the same general formula.

- a. Mention four characteristics of homologous series. [2]
- b. Write the structure of first and second member of aliphatic ketone. [1]
- c. Define the following with suitable examples. [2]
 - i. Heterocyclic compound
 - ii. Reforming process

OR

- a. Define pyrolysis. Write the possible products of cracking of n-hexane. [2]
- b. Write the IUPAC name of following compounds. [3]



17. Oxides are binary compounds of oxygen with another element except noble gases and noble metals. Some examples are:

ZnO	CO	MgO	P_2O_5	K_2O_2	Pb_3O_4
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- a. Identify the peroxide and mixed oxide from above table. [1]
- b. Which oxide from the above table is amphoteric and which is acidic? Give reasons with suitable chemical reactions. [2]
- c. Mention two industrial applications of oxygen. [1]
- d. What happens when ozone gas is passed into the KI and starch solution. [1]

18. Originally nitric acid was manufactured from sparked N_2 and O_2 in Birkeland - Eyde process. Nowadays, it is manufactured by Ostwald's process.

- a. Write the principle and flow sheet diagram for the manufacture of nitric acid by Ostwald's process. [3]
- b. Why is conc. nitric acid stored in dark brown coloured bottle? [1]
- c. Write the action of conc. HNO_3 with white phosphorus. [1]

19. Sulphuric acid, commonly known as King of chemical / oil of vitriol is widely used in chemical industries for various purpose.

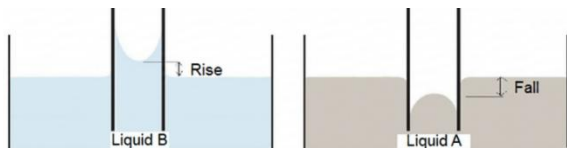
- a. Draw the flowsheet diagram for the manufacture of sulphuric acid by contact process. [2]
- b. Dilution of concentrated sulphuric acid is done by adding concentrated H_2SO_4 into water but not water into sulphuric acid. Why? [1]
- c. What happens when conc. H_2SO_4 is treated with [2]
 - i. methanoic acid
 - ii. sucrose

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

20. There are three distinct physical states of matter, i.e., solid, liquid, and gas.
- Write the statement and derive Graham's law of diffusion. [3]
 - A mixture of ozone and oxygen containing 20% by volume of ozone diffused through a porous plug in 172 seconds, while the same volume of pure oxygen took 164 seconds to diffuse through the same plug. Calculate the relative density of ozone. [2]
 - Two capillary tubes of the same diameter are dipped separately in water and mercury, as shown in the figure. One liquid rises while the other shows a depression. Identify liquid A and liquid B, and explain the behaviour using cohesive and adhesive forces. [2]



- A sample of calcium chloride (CaCl_2) is left exposed to moist air. After some time, it absorbs moisture and turns into a liquid. Which phenomenon explains this behaviour, and why does CaCl_2 not merely become wet but completely liquefy? [1]
21. Answer the followings;
- Write the postulates of VSEPR theory. [3]
 - Why are metals malleable and ductile? [2]
 - What is dipole moment? Write its unit. [1]
 - What is bond length? Mention the factors affecting for bond length. [2]
22. One of the compounds of Sulphur, Hydrogen Sulphide, is used as an analytical reagent for the detection of salt during salt analysis in the lab.
- Write the chemical equation for the preparation of H_2S gas and discuss the working principle of Kipp's apparatus with figure. [1+3]
 - Show that H_2S is acidic in nature. [1]
 - Write the reactions to show the role of H_2S for the precipitation of group II and group IIIB metal ions. [2]
 - What happens when a moist red flower is introduced into a jar containing SO_2 gas? [1]

OR

- Write one chemical reaction to show that nascent hydrogen is better reducing agents than molecular hydrogen. [1]
- List out the isotopes of hydrogen with their number of nucleons and one use of each. [2]
- Why is hydrogen fuel regarded as zero emission fuel? [1]
- What is ring test? Write the chemical equation. [2]
- Why does SO_2 act as both oxidizing and reducing agent, but H_2S only acts as reducing agent? [2]